

# ALEXANDER REEVES

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## EDUCATION

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### **PhD, ETH Zürich**

*September 2021 - current*

PhD candidate in Professor Alexandre Refregier's Cosmology group with secondary supervision from Professor Andrina Nicola (Universität Bonn).

Multiprobe analysis combining Cosmic Microwave Background and Large Scale Structure data.

Using the Effective Field Theory of Large Scale Structure (EFTofLSS) to analyse galaxy clustering.

### **Masters, Downing College, University of Cambridge**

*September 2020 - July 2021*

Masters: Part III Natural sciences (Astrophysics)

Masters grade: 1<sup>st</sup> class (ranked 4<sup>th</sup> in cohort)

### **Bachelors, Downing College, University of Cambridge**

*September 2017 - July 2020*

Undergraduate: Part II Physics

Undergraduate grade: 1<sup>st</sup> class

## AWARDS

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### **Doc.Mobility fellowship, ETHZ**

*2024*

Awarded (9100CHF) fellowship to undertake research at the University of California, Berkeley and Lawrence Berkeley National lab under the supervision of Dr.Simone Ferraro and Professor Martin White.

### **1st place: International PLANCKS theoretical Physics competition**

*2020*

Part of the 4-person University of Cambridge team that came first out of 45 University teams at the international finals.

### **Whitelegg and Unwin Scholarships, Downing College**

*2018 & 2021*

Scholarships for academic achievement.

## STUDENT SUPERVISION

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### **Domitille Chebat, Master thesis co-supervisor**

*February 2024 - September 2024*

Thesis title: 'Realistic field level simulations with the effective field Theory of Large Scale Structure'

### **Laura Reymond, Master thesis co-supervisor**

*February 2024 - September 2024*

Thesis title: 'Precise computation of the angular power spectrum at large angular scales'

### **Laura Reymond, Semester thesis supervisor**

*September 2023 - December 2023*

Thesis title: 'Precise computation of the angular power spectrum at large angular scales'

### **Fabio Holdener, Master thesis co-supervisor**

*February 2023 - August 2023*

Thesis title: 'Weak Lensing With 21cm Intensity Mapping'

### **Luca Morf, Master thesis co-supervisor**

*September 2022 - March 2023*

Thesis title: 'Field-level Simulations from the Effective Field Theory of Large-Scale Structures'

### **Peter Bajzath, Semester project supervisor**

*September 2022 - December 2022*

Thesis title: 'Investigating measurement tensions with Machine Learning emulators'

### **Luis Machado Poletti Valle, Semester project supervisor**

*February 2022 - May 2022*

Thesis title: 'The Integrated Sachs-Wolfe effect on the lightcone'

## RELEVANT TEACHING EXPERIENCE

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Teaching assistant for ‘Astrophysics II’  
Teaching assistant for ‘Cosmological probes’  
Teaching assistant for ‘Theoretical cosmology’

*February 2024 - September 2024*  
*September 2023 - December 2023*  
*February 2023 - May 2023*

## SELECTED TALKS AND POSTERS

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Invited talk, Tensions in Cosmology  
**Invited conference talk based on arXiv:2502.01722**

*Corfu, September 2025*

Invited talk, Recontres du Vietnam  
**Invited conference talk based on arXiv:2502.01722**

*Quy Nhon, August 2025*

INPA colloquium, Lawrence Berkeley National Laboratory  
**Colloquium talk based on arXiv:2502.01722**

*Berkeley, February 2025*

Tea leaks talk at Stanford University  
**Short talk based on arXiv:2502.01722**

*Stanford, January 2025*

Tensions in Cosmology  
**Delivered a 10-minute talk on results of the paper: arXiv:2207.01501**

*Corfu, August 2022*

Talk at NBI Summer school: ‘Neutrinos here there and everywhere’  
**Delivered a 10-minute talk on results of the paper: arXiv:2207.01501: link**

*Copenhagen, June 2022*

## COLLABORATION MEMBERSHIPS

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Dark Energy Spectroscopic Instrument (DESI) member  
Legacy Survey of Space and Time-Dark Energy Science Collaboration (LSST-DESC) member

## PUBLICATIONS

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1. **A.Reeves**, P. Zhang, H. Zheng, *Debiasing inference in large-scale structure with non-flat volume measures*, July 2025, submitted to JCAP, arXiv:2507.20990  
**Contribution: computational analysis, paper writing.**
2. **A.Reeves**, P. Zhang, H. Zheng, *PyBird-JAX: Accelerated inference in large-scale structure with model-independent emulation of one-loop galaxy power spectra*, July 2025, submitted to JCAP, arXiv:2507.20990  
**Contribution: computational analysis lead and paper writing.**
3. L.Reymond, **A.Reeves**, P.Zhang, A.Refregier, *SwiftC<sub>ℓ</sub> fast differentiable angular power spectra beyond Limber*, May 2025, submitted to JCAP, arXiv:2504.01669  
**Contribution: concept, supervision, paper editing.**
4. Di Valentino et al. (including **A.Reeves**), *The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics*, April 2025, arXiv:2504.01669  
**Contribution: wrote part of the ‘Early dark energy and variants’ section**
5. **A.Reeves**, A.Nicola A.Refregier, *Tuning the cosmic instrument: robust cosmology through combined probes*, February 2025, accepted in JCAP, arXiv:2502.01722  
**Contribution: lead the project and performed the computational analysis and majority of paper writing.**
6. **A.Reeves**, A.Nicola A.Refregier, T.Kacprzak, L.F.Machado Poletti Valle, *12×2pt combined probes: pipeline, neutrino mass, and data compression*, January 2024, accepted in JCAP, arXiv:2309.03258.  
**Contribution: led the project and performed the majority of the computational analysis and subsequent writing of the paper.**

7. **A.Reeves**, L.Herold, S.Vagnozzi, B.Sherwin, E.Ferreira, *Restoring concordance with Early Dark Energy and Massive neutrinos*, April 2023, accepted in MNRAS, arXiv:2207.01501.

**Contribution:** led the project and performed the bulk of the theoretical and computational analysis as well as subsequent writing of the paper.

## SOFTWARE

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1. Led the development of **UFalcon2** a software tool in **Python** for rapidly post-processing N-body code output into signal maps for several cosmological probes source code, documentation.
2. Leading the development of **PyBird-JAX** a software tool in **Python-JAX** for rapid, differentiable EFTofLSS predictions.